

FINAL EXPLANATION: CHEMISTRY GRADE 10

Student Assessment Document

FINAL EXPLANATION: CHEMISTRY GRADE 10 – SUB-STRAND 1.2 THE ATOM: FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

Your driving question is: Why do different substances produce different colours when burned in a flame, and what does this reveal about how electrons are arranged inside an atom? Write a clear explanation in the blank space under each prompt. Use evidence from your lessons: the flame tests, the gold foil simulation, your Bohr and orbital drawings, the isotope and relative atomic mass calculations, and your electron configurations. Refer to the Rift Valley cooking hearth and the Kisumu school laboratory where you can. Write in full sentences, use correct chemistry words, and show any working. Your teacher will mark your answer against the rubric.

Part 1: The Puzzle and What Atoms Are Made Of

Describe what you saw at the start of the unit when salt, banana-peel ash and different substances were put in a flame. Why did this seem puzzling when you knew that all substances are made of atoms and fire is just heat? Then use the Rutherford gold foil simulation to explain how scientists learned that the atom has a nucleus. For each of the three observations (most particles, a small number, a very small number), say what it told scientists about the atom. Then say what the Rutherford model could not yet explain about the flame colours.

Write your answer here:

Part 2: Energy Levels and the Origin of Flame Colour

Using Bohr's model, explain step by step what happens to the outermost electron of a sodium atom when salt falls into the cooking fire. Say how many electrons sodium has in each shell and write its electron arrangement. Explain why the light that comes out is a particular colour, and why potassium gives a different colour from sodium.

Write your answer here:

Part 3: Orbitals and Electron Configuration

(a) Explain why the Bohr shell model was not enough and what s and p orbitals add. (b) State the three rules that decide how electrons fill orbitals. (c) Write the s/p electron configuration of sodium and of potassium, and explain how the configurations help to explain the different flame colours of sodium and of potassium.

Write your answer here:

Part 4: Isotopes and Relative Atomic Mass

Chlorine atoms from the ocean near Mombasa do not all have the same mass. Data card: Cl-35 (75%) and Cl-37 (25%). (a) Explain what isotopes are and how the two chlorine forms on the card differ. (b) Calculate the relative atomic mass of chlorine and show a reasonableness check. (c) Would Cl-35 and Cl-37 give different flame colours? Explain your answer using electrons.

Write your answer here:

Part 5: Answering the Driving Question

Write a complete answer to the driving question: Why do different substances produce different colours when burned in a flame, and what does this reveal about how electrons are arranged inside an atom? Use the cooking hearth in the Rift Valley as your example, and use evidence from across the unit to support your answer.

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Use of evidence from the lessons	Uses specific evidence from several lessons, such as the flame test colours, gold foil results, Bohr diagrams, orbital configurations and the chlorine data, and links each piece to the explanation.	Uses some evidence from the lessons, but the link to the explanation is sometimes missing or general.	Gives little or no evidence, or states facts without any link to the lessons.
Explanation of electron energy transitions	Correctly sequences absorption of energy, jump to the excited state, fall to a lower level and emission of a photon, and explains that the colour depends on the energy gap, which is unique to each element.	Describes the jump and the release of light but is vague about why the colour is specific to the element.	Describes the flame colour without a correct explanation of electron jumps or energy release.
Electron arrangement and configuration	Writes correct configurations for sodium and potassium, applies Aufbau, Pauli and Hund rules, and explains how $3s^1$ versus $4s^1$ leads to different colours.	Writes mostly correct configurations and states the rules, with limited link to the colours.	Configurations are incorrect or missing, and the rules are confused.
Isotopes and relative atomic mass	Calculates 35.5 correctly with a reasonableness check and explains that isotopes share electron arrangement and so the same flame colour.	Calculates correctly but gives an incomplete explanation of the flame colour of isotopes.	Calculation is incorrect or the idea of isotopes is confused.
Scientific communication	Uses precise vocabulary (nucleus, shell, orbital, excited state, photon), a logical order and the Kenyan context throughout.	Uses mostly correct vocabulary and a clear order, with minor errors.	Uses little scientific vocabulary and the ideas are hard to follow.

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Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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